Meet Dhamecha

PYTHON FULL-STACK DEVELOPER
AI / LLM ENGINEER

✦ 2+ Years Experience

CONTACT

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HIGHLIGHTS

- **14** Production AI & backend projects delivered
- **2+** Years professional experience
- **500K+** Records scraped & processed weekly
- **30K+** Lines of code — enterprise platform
- **80%** LLM API cost reduction via local models
- **100×** Query speedup with Redis caching

TECHNICAL SKILLS

LANGUAGES & BACKEND

Python · FastAPI · Django · DRF · Flask · SQLAlchemy · REST API · Pydantic

AI / LLM ENGINEERING

LangChain · LangGraph · Ollama · Gemini API · Claude API · HuggingFace · MCP · AI Agents · Tool Calling · Prompt Eng.

RAG & VECTOR DATABASES

RAG Pipelines · CAG · Qdrant · ChromaDB · PGVector · Embeddings · Semantic Search · HNSW

DATA SCIENCE & ML

Pandas · NumPy · Scikit-learn · XGBoost · Matplotlib · Seaborn · Plotly · SHAP · EDA

SCRAPING & AUTOMATION

Selenium · BeautifulSoup · XPath/CSS · n8n · Headless

DATABASES

PostgreSQL · MySQL · SQLite · Redis · Alembic

DEVOPS & CLOUD

Docker · Kubernetes · AWS EC2/S3/RDS

Professional Summary

Results-driven **Python Backend & AI Systems Developer** with **2+ years** of hands-on experience building scalable REST APIs, intelligent automation pipelines, and production-grade LLM-powered applications. Proficient in **Django, FastAPI**, and full-stack data workflows using Pandas and NumPy. Expert in modern AI engineering with **LangChain, LangGraph, Hugging Face Transformers, Ollama, and MCP (Model Context Protocol)**, with deep specialization in designing **Retrieval-Augmented Generation (RAG) and Cache-Augmented Generation (CAG) systems**, autonomous AI agent frameworks, and document intelligence platforms backed by Qdrant, ChromaDB, and PGVector. Proven ability to integrate multiple LLM providers (Gemini, Claude, OpenRouter, HuggingFace) with automatic failover and cost optimization. Comfortable shipping containerized microservices with **Docker and Kubernetes**, building CI/CD pipelines, and deploying to **AWS** infrastructure. Passionate about building intelligent, performant, and maintainable AI products that solve real-world problems.

RAG / CAG Systems · AI Agents + MCP · LangChain / LangGraph · FastAPI / Django · Qdrant / ChromaDB · Docker / AWS · Multi-LLM Integration · WebSocket / Real-time

Professional Experience

Backend Developer
TeamOpine Solutions — Surat, Gujarat · Jan 2025 – Present

Stack: Python · FastAPI · LangChain · LangGraph · RAG · CAG · MCP (Model Context Protocol) · Celery · Redis · PostgreSQL · Docker · Gemini API · Claude API · HuggingFace · Qdrant · ChromaDB

- Developed **high-performance FastAPI microservices** with role-based authentication, JWT security, input validation via Pydantic, comprehensive OpenAPI/Swagger documentation, and Celery + Redis background task processing for heavy ETL operations, file processing, and bulk email dispatch — reducing endpoint response times by over **60%**.
- Designed and deployed a **multi-document RAG system** supporting PDF, DOCX, XLSX, CSV, and image inputs; engineered the full ingestion pipeline including format-specific text extraction (PyMuPDF, python-docx, Pandas, Tesseract OCR), configurable chunking with overlap, embedding generation via Hugging Face sentence transformers, and hybrid vector search (dense + sparse) using both Qdrant and ChromaDB.
- Architected a **production-grade RAG + CAG chatbot platform** with real-time WebSocket communication, Redis pub/sub for event distribution, and a dual-layer semantic caching system achieving **sub-100ms** cached answer retrieval (5–10ms) and **sub-500ms** full pipeline latency; deployed with 5-microservice Docker Compose (FastAPI, Celery workers, Redis, Qdrant, Flower monitoring).
- Built **MCP (Model Context Protocol) servers** using FastMCP framework integrating Gemini 2.5-Flash and Anthropic Claude API with file management tools, response persistence, and agent-to-service communication; developed multi-LLM AI agent frameworks unifying OpenRouter, HuggingFace (DeepSeek-V3.2), Gemini, and Claude under a single abstraction layer.
- Architected a **graph-based multi-agent automation framework** using LangGraph and LangChain: Orchestrator agents for task routing, SQL Executor agents for NL-to-SQL translation, Chart Generation agents, and Monitor agents for health checks — all connected to WhatsApp and Telegram bots; used Ollama to run Llama 3 and Mistral locally, achieving **80% reduction in LLM API spend**.
- Integrated **Gemini 2.0 Flash Vision** and Hugging Face vision models to power an AI food nutrition prediction backend serving an Android application — processing 1,000+ daily requests at sub-800ms average response time; collaborated on **Power BI dashboards** and analytics pipelines using Pandas, NumPy, and Plotly for business reporting.

Django Developer
Zerobit Infotech — Surat, Gujarat · Jul 2024 – Dec 2024

GitHub Actions · GitLab CI · Nginx · Gunicorn

BI, TOOLS & PRACTICES

Power BI · DAX · Chart.js · D3.js · Git · Celery · WebSocket · Postman · Pytest

EDUCATION

Bachelor of Science in Information Technology
J.P. Dawer Institute of Information Science & Technology
Veer Narmad South Gujarat University, Surat
2020 – 2024

CORE STRENGTHS

- Deep LLM internals: tokenization, context windows, attention mechanisms & advanced prompt engineering for open-source and proprietary models
- MCP (Model Context Protocol) server architecture — tool design & AI agent-to-service integration
- Vector search systems: embedding model selection, hybrid retrieval strategies, chunk sizing & re-ranking
- Production AI safety: rate limiting, prompt injection prevention, SQL injection protection & secure API key management
- Full DevOps lifecycle: multi-service Docker, CI/CD pipelines, AWS IAM & security configuration

Stack: Python · Django · Django REST Framework · PostgreSQL · JWT · Redis · Celery · Ubuntu Server · Nginx · Gunicorn · Git

- Built and maintained **scalable Django and DRF web applications** implementing modular architectures with JWT authentication, DTO-based request handling, schema validation layers using DRF serializers, and optimized PostgreSQL integration to support complex multi-tenant business workflows.
- Designed and implemented **RESTful APIs using Django REST Framework** with serializers, viewsets, routers, custom permissions, and pagination — supporting both internal services and third-party client integrations with comprehensive OpenAPI documentation.
- Developed **reusable Django apps and middleware** for authentication, request logging, and input validation; integrated Celery + Redis for background task processing including report generation, email dispatch, and scheduled data pipeline jobs.
- Deployed production Django services on **Ubuntu servers** — configured Nginx reverse proxy, Gunicorn WSGI server, environment-based settings, static file serving, PostgreSQL connection pooling, and performed performance tuning and live troubleshooting.

Key Projects 14 Projects

PROJECT 01

MCP RAG + CAG Chatbot Platform

FastAPI · LangChain · Qdrant · Redis · Celery · WebSocket · Docker · Gemini · Claude API · Sentence Transformers

- Architected a **production-grade RAG + CAG (Cache Augmented Generation)** chatbot platform with real-time WebSocket communication and 5-microservice Docker Compose orchestration — FastAPI (API), Celery (async workers), Redis (pub/sub + caching), Qdrant (vector DB), Flower (monitoring dashboard).
- Implemented **MCP server** with Gemini 2.5-Flash + Anthropic Claude API integration providing 4 tools: ask_gemini, save_response, read_response, list_responses; unified multi-LLM interface across Gemini, Claude, HuggingFace DeepSeek-V3.2, and OpenRouter.
- Dual-layer caching: **Redis semantic answer cache** (92% cosine similarity threshold) delivers 5–10ms hits; full RAG pipeline (Qdrant HNSW search + Gemini generation) completes in 400–500ms; average latency **~250ms** with caching; throughput scales to **32–40 questions/sec** with 4 workers.
- Real-time frontend UI: WebSocket connection, Markdown rendering (marked.js + DOMPurify XSS sanitization), source attribution with relevance scores + page numbers, typing indicators, latency metrics display, cache hit badges, and streaming token-by-token responses.
- Document ingestion pipeline for PDF, DOCX, TXT, JSON — automatic chunking (512 tokens, 50 overlap), Sentence Transformer embedding, Qdrant upsert with metadata; metadata filtering by category in vector search queries.

PROJECT 02

Database Insight Assistant — Enterprise AI Analytics

Django DRF · LangChain · Qdrant · Redis · Celery · Chart.js · D3.js · TailwindCSS · Docker · 4 LLM Providers

- Full-stack AI-powered database analytics platform — **60+ REST endpoints, 77 API view classes, 7 Celery tasks, 12 HTML templates, 54 Python files, ~30,000+ LOC**; glassmorphism frontend with D3.js force-directed relationship graphs and Chart.js 4.x visualizations.
- **4-agent orchestration system**: SQL Agent (NL-to-SQL with relationship detection), Analyst Agent (statistical analysis + anomaly detection), Reporter Agent (dashboard + report generation), Monitor Agent (data quality profiling); **4 LLM providers** (Gemini, Groq, HuggingFace, OpenRouter) with automatic failover.
- **Power BI-like dashboard engine**: AI single-prompt dashboard generation, drag-and-drop widget builder, 40+ DAX-like measures (aggregations, time intelligence YoY/MoM/YTD, statistical functions), cross-filtering, and portable `.dbia` export format; **100× query speedup** via Redis 256MB LRU cache.
- Cross-database support — SQLite, PostgreSQL, MySQL, MSSQL; file uploads (SQLite, CSV, JSON, XML, SQL dumps); WAL mode + busy timeout + exponential backoff for database lock resolution; SQL injection protection via parameterized queries; Docker Compose (5 services: Django, Redis, Qdrant, Celery, Flower).

PROJECT 03

AI Agents with MCP Framework

MCP · FastMCP · LangChain · OpenRouter · Gemini · HuggingFace DeepSeek-V3.2 · SQLAlchemy · Rich CLI

- Multi-module AI agent framework demonstrating the **LLM vs. Agent paradigm distinction** — SimpleLLM (intelligence only) vs. AIAgent (decision-making, task prioritization, state persistence, action execution in external systems).
- **FastMCP server** with 4 tools: ask_gemini, save_response, read_response, list_responses; integrates Gemini 2.5-Flash model; standardized MCP tool interface for AI-to-service communication.
- **DatabaseAIAgent** with multi-database connectivity (MySQL, PostgreSQL, MSSQL via SQLAlchemy); natural language schema understanding, NL-to-SQL generation, safe query execution, and results formatted via Rich terminal UI with formatted tables and panels.
- HuggingFace DeepSeek-V3.2 integration with both standard and **streaming token-by-token responses**; environment-based configuration; comprehensive test suite for MCP tools.

PROJECT 04

Document Intelligence System — Multi-Format RAG

LangChain · Qdrant · ChromaDB · PGVector · HuggingFace · FastAPI · Ollama · Tesseract OCR · PyMuPDF

- Full document intelligence platform — users upload **PDF, DOCX, XLSX, Excel, and image files** and have natural-language conversations with their content powered by a complete RAG pipeline with configurable chunk overlap and dynamic metadata tagging.
- Format-specific extractors: PyMuPDF (PDF), python-docx (DOCX), Pandas (Excel/XLSX), **Tesseract OCR** (images); HuggingFace sentence transformer embeddings indexed in both Qdrant and PGVector for redundancy.
- **Hybrid search** (dense + sparse retrieval) improving recall for technical and factual queries; LangChain context-window compression, relevance-ranked chunk injection, and structured prompts for accurate grounded responses.
- **Multi-tenant document isolation** — each user's documents and conversation history remain private; Ollama-backed local LLM option (Llama 3, Mistral) for enterprise clients with on-premise data requirements and air-gapped environments.

Key Projects — Continued

Projects 05–10

PROJECT 05

Database Communicator RAG

Gemini API · ChromaDB · Sentence Transformers · SQLAlchemy · MySQL · Pandas · Python

- Production-ready **RAG-based NL-to-SQL system** converting plain-English questions into optimized SQL queries and returning answers in natural language; multi-stage pipeline: Cache Check → Schema Retrieval → SQL Generation → Execution → NL Synthesis → Cache Store.
- **Semantic similarity caching** using ChromaDB embeddings with configurable threshold (default 0.88); finds similar previous questions, returns cached answer at ~5ms — eliminates redundant LLM API calls, significantly reducing cost.
- Dynamic schema population into vector store — table/column relationships stored as embeddings for context-aware SQL generation; schema-aware prompting reduces hallucinations vs. naive approaches.
- **Auto CSV export** for large datasets (configurable row threshold, default 50); NL answer synthesis generating conversational responses with insights from query results; cache hit/miss statistics tracking and database summarization.

PROJECT 06

Multi-Agent AI Automation Platform

LangGraph · LangChain · Ollama · FastAPI · WhatsApp API · Telegram Bot API · PostgreSQL

- **Graph-based multi-agent framework** using LangGraph with 4 distinct node agents: Orchestrator Agent (intent routing and task delegation), Query Agent (NL-to-SQL translation), Chart Maker Agent (visualization generation), Monitor Agent (health checks and anomaly alerting).
- WhatsApp and Telegram automation bots connected to the agent framework — users send commands like *"show this week's sales by region"* and receive **chart images directly in the chat**; supports structured queries, report generation, and CSV attachments.
- Database Communicator Agent — translates free-text questions to SQL, executes safely against live PostgreSQL with parameterized queries, formats results into readable summaries or downloadable CSV files.
- Ollama-hosted **local LLMs (Llama 3, Mistral)** for cost-sensitive agent tasks with graceful fallback to Gemini API for complex reasoning — achieving **80% LLM API spend reduction**; fault-tolerant retry policies, agent state persistence, and full audit logging across all workflow nodes.

PROJECT 07

AI Food Nutrition Analyzer

FastAPI · Gemini 2.0 Flash Vision · MySQL · SQLAlchemy · AWS EC2 · AWS RDS · PIL/Pillow · Uvicorn

- Complete backend for an **Android application** — accepts food images via multipart upload, analyzes using Gemini 2.0 Flash Vision AI, returns detailed per-dish nutrition breakdowns: calories, protein, carbohydrates, fat, fiber, sugar, sodium, cholesterol, count of servings, and meal-time recommendations (breakfast/lunch/dinner/snack).
- Serves **1,000+ daily requests** with sub-800ms average response time on AWS EC2 + RDS PostgreSQL; image optimization pipeline (PIL resizing, base64 encoding) keeps per-request token cost within budget while maintaining prediction accuracy above 90%.
- 3 REST endpoints: POST /api/analyze-food (upload + analyze), GET /api/analyze-food (paginated history with date-range filtering), POST /api/user-register (device registration); **API key authentication** via x-api-key header, CORS support, UUID-based secure file storage.
- **Token usage tracking** (input/output/total per request) for cost monitoring; SQLAlchemy ORM with 2 tables (food_analyses, user); input sanitization + rate limiting middleware preventing prompt injection attacks and unauthorized access.

PROJECT 08

Machine Learning Prediction & Analytics

Scikit-learn · XGBoost · Pandas · NumPy · SHAP · Matplotlib · Seaborn · Plotly · FastAPI · Jupyter

- Trained and evaluated multiple **classification and regression models** — Random Forest, XGBoost, Logistic Regression — across business datasets including customer churn prediction, product price estimation, and demand forecasting.
- End-to-end ML workflows: **EDA, feature engineering** (encoding, scaling, interaction features), hyperparameter tuning with GridSearchCV + RandomizedSearchCV, k-fold cross-validation, and **SHAP value interpretability reports** explaining model decisions to non-technical stakeholders.
- Served trained models as **FastAPI REST inference endpoints** — enabling real-time predictions integrated into frontend dashboards and downstream business applications without exposing raw model code.
- Packaged analytical pipelines into **interactive HTML dashboards** (Plotly) and **automated PDF reports** (Matplotlib) delivered to stakeholders, fully replacing manual Excel-based reporting processes.

PROJECT 09

Power BI Analytics Dashboards

Power BI · DAX · Python · PostgreSQL · Pandas · NumPy · Direct Query

- **College attendance tracking dashboard** with department-wise, student-wise, and date-range views; drill-through capabilities enabling administrators to navigate from summary KPIs down to individual student attendance records.
- **Inventory management dashboard** for a manufacturing client — live PostgreSQL Direct Query connection, DAX-calculated KPIs: turnover rate, reorder level alerts, stockout risk scores, supplier performance metrics; **automated email alerts** triggered on threshold breaches.
- Pre-processed raw data with Python and Pandas before loading into Power BI data models — data cleaning, type normalization, calculated columns — significantly improving query performance and reducing dashboard load times.

PROJECT 10

Secure Microservice API with Async Task Processing

FastAPI · Celery · Redis · SQLAlchemy · PostgreSQL · MySQL · Alembic · Pytest · GitLab CI · Postman

- Production **FastAPI microservice** with layered security: OAuth2 + JWT tokens, refresh token rotation, IP-based rate limiting, and role-based permission guards on all resource endpoints; Celery + Redis broker offloaded heavy operations (PDF generation, bulk email dispatch, data transformation) — reducing endpoint response times by **over 60%**.
- Abstracted database with **SQLAlchemy ORM + Alembic migrations** supporting dual-database environments (MySQL for legacy integration, PostgreSQL for new services) with zero-downtime schema changes.
- Comprehensive **Pytest test suite achieving 85%+ code coverage**; automated API contract validation using Postman collections; integrated into GitLab CI pipeline with automated test runs on every push.
- Detailed **OpenAPI/Swagger documentation** with request/response schemas, authentication flows, and error-code references; significantly reduced frontend integration time and onboarding for new team members.

Key Projects — Continued

Projects 11–14

PROJECT 11

Full-Stack E-Commerce with Finance Modules

Django · PostgreSQL · Pandas · OpenPyXL · Nginx · Gunicorn · Ubuntu · select_related

- Production-grade **e-commerce application from scratch** — product catalog management, shopping cart, order processing, role-based admin dashboard with full CRUD; deployed on Ubuntu with Nginx + Gunicorn and PostgreSQL connection pooling.
- Complex **finance calculation modules**: GST-inclusive invoicing with tax slab logic, profit/margin tracking per product and order, monthly/quarterly revenue summaries; reduced manual accounting effort for the business client by approximately **70%**.
- Automated **Excel report exporter** using OpenPyXL and Pandas — generates downloadable invoices, stock reports, and sales analytics sheets on demand or via scheduled triggers; reports include formatted cells, formulas, and charts.
- Database optimization: added indexes, applied **select_related/prefetch_related** to eliminate N+1 queries, query plan analysis — reduced average API response time from **820ms to under 190ms** for product listing endpoints (4.3× speedup).

PROJECT 12

Multi-Site Web Scraping & Inventory Automation

Selenium · BeautifulSoup · Pandas · n8n · Cron · SQLite · Proxy Rotation · Stealth Plugins · JSON

- Engineered an automated scraping framework targeting **12+ US-based e-commerce and inventory management portals** with dynamic JavaScript-rendered content; used Selenium with rotating user agents, stealth plugins, proxy rotation, and CAPTCHA-handling strategies — maintaining **95%+ uptime** on long-running headless Ubuntu servers.
- Scraped **500,000+ product and inventory records weekly**; applied Pandas normalization pipelines to clean, deduplicate, and standardize data, then exported structured CSVs for downstream business reporting and client consumption.
- Built a **cultural data scraping collection**: 250K+ Gujarati wisdom quotes, 700+ Bhagavad Gita shlokas (Sanskrit + commentary), yoga asanas + mudras, astrology forecasts, and Gujarati proverbs/riddles from 10+ websites — full Unicode support (Gujarati, Marathi, Hindi); data stored in JSON and SQLite.
- Orchestrated scheduling, alerting, and pipeline handoff using **n8n workflows**; retry logic, stateful pagination with back-button navigation, JavaScript DOM execution for dynamic content and paywall bypass.

PROJECT 13

Amazon Scraper — End-to-End ETL Pipeline

Selenium · Pandas · MySQL · mysql-connector-python · WebDriverWait · Jupyter Notebook · CSV

- Selenium-based **Amazon.in laptop scraper** with multi-page pagination (JavaScript-click "Next Page"), scroll-to-load dynamic content, stale element recovery (StaleElementReferenceException handling), explicit WebDriverWait for element readiness, and NoSuchElementException graceful fallback to "N/A".
- Extracted **64KB+ structured product dataset** — product name, current price, discount percentage, original MRP — across multiple pages covering 100+ laptop listings from multiple brands and price ranges.
- Complete **ETL pipeline**: Selenium scrape → CSV export → Pandas DataFrame transformation (NaN handling, type conversion, data validation) → MySQL batch insertion with transaction management and connection pooling.
- Exploratory data analysis and visualization via **Jupyter Notebook** on the scraped dataset and Pokemon dataset — descriptive statistics, correlation analysis, bar/scatter/box/heatmap plots using Matplotlib and Seaborn.

PROJECT 14

Pandas Data Visualization & EDA

Pandas · NumPy · Matplotlib · Seaborn · SciPy · Jupyter Notebook · CSV · Statistical Analysis

- Comprehensive Jupyter Notebook (~533KB) demonstrating **end-to-end Exploratory Data Analysis** workflow on the Ice Cream Ratings dataset — data loading, structural inspection (.head, .info, .describe), missing value detection and treatment, duplicate removal, and type normalization.
- **Advanced Pandas operations**: groupby aggregations, pivot tables, multi-column merge/join, conditional filtering, Boolean indexing, computed columns, rolling windows, and statistical operations (mean, median, std, corr, quantiles).
- Multi-dimensional visualizations using Matplotlib and Seaborn: **univariate** (histograms, KDE, box plots), **bivariate** (scatter with regression, joint plots), **multivariate** (heatmaps, pair plots, facet grids, color-encoded scatter) — all with professional styling, labels, and annotations.
- Statistical analysis: Pearson/Spearman correlation matrices, distribution fitting, outlier detection via IQR method, and actionable insight extraction demonstrating **data storytelling** best practices for communicating findings to business stakeholders.

What I Bring to Your Team

AI Engineering Depth

Deep understanding of LLM internals — tokenization, context windows, attention mechanisms, and RAG/CAG architecture design. Proven ability to integrate Gemini, Claude, Ollama, and HuggingFace models into production systems with cost optimization strategies (80% API spend reduction achieved).

Backend & System Design

Comfortable architecting complex distributed systems — async microservices, Redis pub/sub, Celery task queues, WebSocket real-time comms. Strong focus on performance optimization: 4.3× API speedup via query optimization, 100× speedup via intelligent caching layers.

Data & Analytics

Full data pipeline ownership — collection (Selenium, 500K+ records/week), transformation (Pandas, ETL), storage (PostgreSQL, vector DBs), analysis (EDA, ML, SHAP), and visualization (Power BI, Plotly dashboards). Experienced replacing manual Excel workflows with automated pipelines.

Technologies at a Glance

LLM PROVIDERS

Gemini 2.0/2.5 Flash
Anthropic Claude
Llama 3 (Ollama)
Mistral (Ollama)
DeepSeek-V3.2

AI FRAMEWORKS

LangChain
LangGraph
MCP / FastMCP
Hugging Face Transformers
Sentence Transformers

BACKEND & APIS

FastAPI + Pydantic
Django + DRF
Flask
SQLAlchemy + Alembic
Celery + Redis

INFRA & CLOUD

Docker + Compose
Kubernetes
AWS EC2 / S3 / RDS
GitHub Actions
GitLab CI

OpenRouter API
HuggingFace Inf. API

Ollama
Tiktoken

WebSocket
OpenAPI/Swagger

Nginx + Gunicorn
Ubuntu Server